

Ciaran Nimick

ciarannimick@gmail.com | (310) 421 6908 | [ciarannimick.com](mailto:ciarannimick@gmail.com) | [linkedin.com/in/ciaran-nimick](https://www.linkedin.com/in/ciaran-nimick)

Professional Summary:

Highly motivated Mechanical Engineering student seeking a Summer 2027 Mechanical Engineering internship, with a proven ability in mechanical design, FEA, and validation gained through Baja SAE and industry experience at Northrop Grumman, using Siemens NX, Ansys, Femap, and SolidWorks.

Education:

Case Western Reserve University - Cleveland, OH

Expected May 2028

Bachelor of Science in Mechanical Engineering | Minor in Supply Chain Management | GPA: 3.4

Relevant Coursework: Statics & Strengths of Materials | Dynamics | Fluid Mechanics | Mechanical Manufacturing

Experience:

Northrop Grumman - Structures Design, Analysis, and Integration Intern

May 2026 - July 2026

- Validated structural integrity of antenna brackets, radiation shields, and bolted joint connections for Space Development Agency satellites using Femap/Nastran, ensuring compliance with margin of safety requirements
- Performed preliminary structural analysis on satellite architecture configurations, identifying design risks that informed design overhaul and supported a reduction in cost & lead time
- Released 4 engineering drawings applying GD&T principles, ensuring manufacturability and compliance with program drawing standards

CWRU Motorsport Baja SAE

September 2024 - Present

- Operated manual machining tools, including lathes, mills, and tube benders, to fabricate over 35 unique steel, aluminum, bronze, and titanium components with tolerances as tight as ± 0.0005 inches

Rear Outboard Analysis Lead

June 2026 - Present

- Designing the team's rear upright in Siemens NX to address manufacturability and performance issues from last year's design; conducting iterative structural FEA in Ansys Mechanical to validate improvements
- Responsible engineer for validation of all rear outboard components, validating various factors of safety (FoS) based on a confidence interval method to withstand fatigue

Suspension Engineer

April 2025 - June 2026

- Designed and validated the team's 2026 front lower suspension arms, overseeing a 10% increase in ride clearance and a 15% weight reduction while improving serviceability.
- Executed Finite Element Analysis (Ansys) on front outboard components in assembly-level analysis to validate structural integrity under maximum static loads, validating the factor of safety based on a confidence interval method to withstand fatigue
- Aided in developing vehicle setups for various racing environments through adjustments to suspension and steering
- Developed a Java-based design tool that produces a chart of feasible tie rod dimensions and graphically visualizes the factor of safety

Case Western Reserve University - Mechanical Manufacturing TA

August 2025 - Present

- Instructed and mentored 64 students in DFMA principles and SolidWorks part design, assemblies, and drawings, preparing them for the Certified SOLIDWORKS Associate exam
- Graded ~180 manual sketching assignments, ensuring parts were fully defined and correct projections were drawn

PVPHS VEX Robotics - President, Project Manager, & Lead Designer

August 2020 - June 2024

- Increased annual funding fourfold, tripled team membership, and doubled the number of competition teams
 - Organized an inaugural robotics event, raising \$8,000 and earning congressional recognition from Congressman Ted Lieu
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Skills:

Design: Siemens NX | SolidWorks | Onshape | Fusion 360 | Ansys | Femap | GD&T

Manufacturing: Mill | Lathe | Waterjet | CNC Router | Laser Cutter | Drill Press | Band Saw | Tube Bender

Software Languages: MatLab | C++ | Java